

Aristotelian Logic

proposition (statement) — Tautology : $p \vee \neg p$
 Predicate — Contradiction : $p \wedge \neg p$

Logical Connectives

- conjunction $p \wedge q$ - disjunction $p \vee q$ - implication $p \Rightarrow q$ ($p \Rightarrow q \equiv \neg p \vee q$)
 $\equiv \neg q \Rightarrow \neg p$ - exclusive or (Xor) : $p \vee q \equiv (p \vee q) \wedge \neg(p \wedge q)$

De Morgan's Laws

$$\neg(p \vee q) \equiv \neg p \wedge \neg q$$

$$\neg(p \wedge q) \equiv \neg p \vee \neg q$$

Modus Ponens $(p \Rightarrow q) \wedge p \Rightarrow q$ Modus Tollens $(p \Rightarrow q) \wedge \neg q \Rightarrow \neg p$

Disjunctive Syllogism

$$(p \vee q) \wedge \neg p \Rightarrow q$$

Hypothetical Syllogism

$$(p \Rightarrow q) \wedge (q \Rightarrow r) \Rightarrow (p \Rightarrow r)$$

Logical Quantifier

- universal : $\forall$ - existential : $\exists$ Set Theory $x \in A$, set without element : $\emptyset$, universe of discourse : Ω

$$A = \{x_1, x_2, \dots, x_n\} = \{x \mid p(x)\}$$

subsets (supersets)

$$A \subseteq B \iff (x \in A) \Rightarrow (x \in B)$$

equivalent

$$A = B \iff (x \in A \Rightarrow x \in B) \wedge (x \in B \Rightarrow x \in A)$$

Complement of A : $A^c = \{x \in \Omega \mid x \notin A\}$ Union : $A \cup B = \{x \in \Omega \mid x \in A \vee x \in B\}$ Intersection : $A \cap B = \{x \in \Omega \mid x \in A \wedge x \in B\}$

difference:

$$A \setminus B = \{x \in A \mid x \notin B\}$$

$$= A \cap B^c = A \setminus (A \cap B)$$

$$\bigcup_{i \in I} A_i = \{x \in \Omega \mid \exists i \in I, x \in A_i\}$$

$$\bigcap_{i \in I} A_i = \{x \in \Omega \mid \forall i \in I, x \in A_i\}$$

Disjoint/Mutually exclusive sets $A_i \cap A_j = \emptyset, \forall i \neq j$ Partition : 1) mutually exclusion $A_i \cap A_j = \emptyset$ 2) $\bigcup_{i \in I} A_i = \Omega$

Element Chasing

$$A = B \iff (A \subseteq B) \wedge (B \subseteq A)$$

Properties of Set operations

Commutativity : $A \cap B = B \cap A$ Associativity : $A \cap (B \cap C) = (A \cap B) \cap C$ Distributivity : $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$ Absorptions : $A \cup (A \cap B) = A$

De Morgan's Laws

$$\left(\bigcup_{i \in I} A_i\right)^c = \bigcap_{i \in I} A_i^c$$

$$\left(\bigcap_{i \in I} A_i\right)^c = \bigcup_{i \in I} A_i^c$$

Cartesian Product : $A \times B = \{(a, b) \mid a \in A, b \in B\}$ Family of Subsets of A / Power Set : 2^A Relations : $K = \{A_i \times B, R \subseteq K\}$ Function : map n-tuple to a unique member of B, $f: A \rightarrow B$

domain, co-domain, Cardinality.

Forward Image of $f: A \rightarrow B$

$$f^*: 2^A \rightarrow 2^B, \forall S \subseteq A (S \in 2^A), f^*(S) = \{y \in B, y = f(x), x \in S\}$$

Inverse Image of $f: A \rightarrow B$

$$f^*: 2^B \rightarrow 2^A, \forall T \subseteq B (T \in 2^B), f^*(T) = \{x \in A, y = f(x), y \in T\}$$

Countably Infinite if A and $\mathbb{N}$ have a same cardinality : $f: A \rightarrow \mathbb{N}$ $\mathbb{N}, \mathbb{Z}, \mathbb{Q}, \mathbb{N} \times \mathbb{N}$, uncountable : $[0, 1], \mathbb{R}, \mathbb{R}^c = \mathbb{R} \setminus \mathbb{Q}$

$$f^*\left(\bigcup_{i \in I} A_i\right) = \bigcup_{i \in I} f^*(A_i), f^*\left(\bigcap_{i \in I} A_i\right) \subseteq \bigcap_{i \in I} f^*(A_i), f^*\left(\bigcup_{i \in I} A_i\right) = \bigcup_{i \in I} f^*(A_i)$$

$$y \in f^*\left(\bigcup_{i \in I} A_i\right) \iff \exists x, x \in \bigcup_{i \in I} A_i, y = f(x) \iff \exists x \exists i \in I, x \in A_i, y = f(x)$$

$$\iff \exists i \in I \exists x, x \in A_i, y = f(x) \iff \exists i \in I y \in f^*(A_i) \iff y \in \bigcup_{i \in I} f^*(A_i)$$

one to one : $x_1 \neq x_2 \Rightarrow f(x_1) \neq f(x_2)$

(injective)

$$|A| \leq |B|$$

onto

$$f^*(A) = B$$

(surjective)

$$|A| \geq |B|$$

one to one correspondence

(bijective)

$$|A| = |B|$$

Limit of a sequence of Sets

$$\lim_{n \rightarrow \infty} A_n = A \text{ iff}$$

$$\forall w \in \Omega \lim_{n \rightarrow \infty} I_{A_n}(w) = I_A(w)$$

$$a) \forall n, A_n \subseteq A_{n+1}$$

$$\lim_{n \rightarrow \infty} A_n = \bigcup_{n \in \mathbb{N}} A_n$$

$$b) \forall n, A_n \supseteq A_{n+1}$$

$$\lim_{n \rightarrow \infty} A_n = \bigcap_{n \in \mathbb{N}} A_n$$

Probability Models probable $\longleftrightarrow$ possible

$P(A) = \frac{n_A}{N}$, relative frequency : $\lim_{N \rightarrow \infty} \frac{n_A}{N}$; countably additive function.

Sample Space : Ω , Event : $E \subseteq \Omega$. probability will be assigned

$P(\emptyset) = 0$ $P(\Omega) = 1$: if Ω is countable always can assign probability to all subsets of Ω

σ -field of events : $\mathcal{F} \subseteq 2^\Omega$

1) $\emptyset \in \mathcal{F}, \Omega \in \mathcal{F}$

2) $A \in \mathcal{F} \Rightarrow A^c \in \mathcal{F}$: closure under complement

3) $A_1, A_2, \dots \in \mathcal{F} \Rightarrow \bigcup_{i=1}^\infty A_i \in \mathcal{F}$: closure under countable union.

$(\Omega, \mathcal{F}, P)$: probability space

define $\begin{cases} B_1 = A_1 \\ B_2 = A_1 \setminus A_2 \\ B_3 = A_2 \setminus (A_1 \cup A_2) \end{cases}$

Lemmas

1) $A_1, A_2, \dots, A_n \in \mathcal{F}, A_i \cap A_j = \emptyset \Rightarrow P(\bigcup_{i=1}^n A_i) = \sum_{i=1}^n P(A_i)$

2) $A \in \mathcal{F} \Rightarrow P(A^c) = 1 - P(A)$

Continuity of Probability Measure $P(\bigcup_{i=1}^\infty A_i) = \lim_{n \rightarrow \infty} P(\bigcup_{i=1}^n A_i)$

a) $\forall i \in \mathbb{N}, A_i \subseteq A_{i+1}, P(\bigcup_{i=1}^\infty A_i) = \lim_{n \rightarrow \infty} P(A_n)$

b) $\forall i \in \mathbb{N}, A_i \supseteq A_{i+1}, P(\bigcap_{i=1}^\infty A_i) = \lim_{n \rightarrow \infty} P(A_n)$

3) $A \subseteq B \Rightarrow P(A) \leq P(B)$

if $A \subseteq B, P(B|A) = \frac{P(B) - P(A)}{P(A)}$

4) $A_1, A_2, \dots \in \mathcal{F} \Rightarrow P(\bigcup_{i=1}^\infty A_i) \leq \sum_{i=1}^\infty P(A_i)$

5) $A_1, A_2, \dots, A_n \in \mathcal{F}$

$P(\bigcup_{i=1}^n A_i) = \sum P(A_i) - \sum_{i < j} P(A_i \cap A_j) + \dots + (-1)^{n+1} P(\bigcap_{i=1}^n A_i)$

Probability Measure (Kolmogorov)

probability measure on $(\Omega, \mathcal{F})$ is a function

$P: \mathcal{F} \rightarrow [0, 1]$

1) $P(\Omega) = 1, P(\emptyset) = 0$

2) $A_1, A_2, \dots \in \mathcal{F}, A_i \cap A_j = \emptyset \forall i \neq j$

$P(\bigcup_{i=1}^\infty A_i) = \sum_{i=1}^\infty P(A_i)$

Conditional Probability $P(A|B) = \frac{P(A \cap B)}{P(B)} = P_B(A)$, satisfied probability axioms

1) $0 \leq P_B(A) \leq 1$ 2) $P_B(\Omega) = 1$ 3) $P_B(\bigcup_{i=1}^\infty A_i) = \sum_{i=1}^\infty P_B(A_i)$

Law of total probability $P(A) = P(A|B)P(B) + P(A|B^c)P(B^c)$

Bayes's Rule : $P(H|E) = \frac{P(E|H)P(H)}{P(E|H)P(H) + P(E|H^c)P(H^c)}$ prior

posterior / likelihood

Multiplication Rule

$P(\bigcap_{i=1}^n A_i) = P(A_1) \prod_{i=2}^n P(A_i | A_1 \cap A_2 \cap \dots \cap A_{i-1})$

Independence $P(A \cap B) = P(A)P(B), P(A|B) = P(A)$

$P(A) = P(A|B)P(B) + P(A|B^c)P(B^c) \Rightarrow P(A) = P(A|B) = P(A|B^c)$
 $= P(A)P(B) + P(A|B^c)(1 - P(B))$

Conditional Independence $P(A \cap B | C) = P(A|C)P(B|C)$

$\{A_i \in \mathcal{F} | i \in I\} \forall J \subseteq I, |J| < \infty$ Pairwise Independence

$P(\bigcap_{i \in J} A_i) = \prod_{i \in J} P(A_i)$

Borel Cantelli Lemmas $A_{i.o.} = \bigcap_{n=1}^\infty \bigcup_{i=n}^\infty A_i$

1) $\sum_{i=1}^\infty P(A_i) < \infty \Rightarrow P(A_{i.o.}) = 0$

Erdős-Rényi $P(A_i) = \frac{1}{L^p}$ $\begin{matrix} p < 1 \rightarrow \infty \\ p > 1 \rightarrow 0 \end{matrix}$

2) $\sum_{i=1}^\infty P(A_i) = \infty$ and A_i 's are independent $\Rightarrow P(A_{i.o.}) = 1$